

Lecture 32 — Infinity

CS 173: Discrete Structures · Infinite sets, Dedekind infinity, Hilbert’s hotel

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1 Recall: $\binom{n}{k}$

From Chapter 6 of the reference, for a set X with $|X| = n$ and $n, k \in \mathbb{N}$,

$$\binom{n}{k} := |\{Z \in \mathcal{P}(X) \mid |Z| = k\}|.$$

That is, $\binom{n}{k}$ — read “ n choose k ” — is the number of subsets of cardinality k taken from a set of cardinality n .

Example. Let $n = 5$, $k = 2$, and take $X = \{a, b, c, d, e\}$, so $|X| = 5$. One particular subset of cardinality 2 is $Z = \{b, c\}$. Counting all such two-element subsets gives

$$\binom{5}{2} = 10.$$

2 What does “infinite” mean?

So far we have only counted finite sets. To do cardinality *at scale*, we need a precise notion of an infinite set.

Definition 2.1 (Infinite set). A set X is **infinite** iff

$$(\forall n \in \mathbb{N})(|X| \neq |[n]|),$$

where $[n] = \{1, 2, \dots, n\}$ is the standard finite set of size n . In words: X is infinite iff its cardinality matches no finite cardinality.

Remark. This is the “negative” definition of infinity — X is infinite precisely when no bijection $X \rightarrow [n]$ exists, for any $n \in \mathbb{N}$. Soon we’ll see a striking “positive” characterization (Dedekind’s), where infiniteness is detected by a strange internal symmetry.

3 Dedekind infinite vs. Dedekind finite

A finite set has the property that no proper subset can match its size — if you remove anything, it gets strictly smaller. Dedekind’s idea is to flip this: an infinite set is one that *can* have a proper subset of the same size.

Definition 3.1 (Dedekind infinite). A set X is **Dedekind infinite** iff

$$(\exists Y)(Y \subseteq X \wedge Y \neq X \wedge |X| = |Y|).$$

That is, X has a proper subset Y with the same cardinality as X itself.

Definition 3.2 (Dedekind finite). Given a set X , we say X is **Dedekind finite** iff

$$(\forall Y)(Y \subseteq X \wedge Y \neq X \Rightarrow |Y| < |X|).$$

Every proper subset is *strictly* smaller.

Remark (Finite sets behave nicely). For finite sets A, B :

$$A \subseteq B \wedge A \neq B \implies |A| < |B| \text{ and hence } |A| \neq |B|.$$

So every finite set is Dedekind finite — this is the intuition the definition is trying to capture.

Caution. The strict inequality $|A| < |B|$ is the right tool only when $A \subseteq B$ and $A \neq B$. Without proper containment, $|A| < |B|$ alone tells you nothing about subset structure (e.g. $\{1, 2\}$ and $\{3, 4, 5\}$).

4 Infinite $\iff$ Dedekind infinite

The two definitions of “infinite” — the negative one (no finite cardinality) and the positive one (a proper subset of the same size) — coincide.

Theorem 4.1 (Equivalence of infinity). A set X is infinite if and only if X is Dedekind infinite.

Remark. We won’t prove the full equivalence today — the $(\Leftarrow)$ direction is straightforward, while $(\Rightarrow)$ requires the Axiom of Choice in the general case. The next two theorems are concrete instances of the Dedekind side: each exhibits an infinite set together with a proper subset of equal size.

5 First example: $|\mathbb{N}^+| = |\mathbb{N}|$

Recall $\mathbb{N} = \{0, 1, 2, \dots\}$ and $\mathbb{N}^+ = \{1, 2, 3, \dots\}$. So $\mathbb{N}^+ \subseteq \mathbb{N}$ and $\mathbb{N}^+ \neq \mathbb{N}$ (since $0 \in \mathbb{N} \setminus \mathbb{N}^+$). Despite this, the two have the same cardinality.

Theorem 5.1. $|\mathbb{N}^+| = |\mathbb{N}|$.

Proof. Recall $|\mathbb{N}^+| = |\mathbb{N}| \iff \exists f$ with $f : \mathbb{N} \rightarrow \mathbb{N}^+$ a bijection. Consider $f : \mathbb{N} \rightarrow \mathbb{N}^+$ given by, for any $n \in \mathbb{N}$,

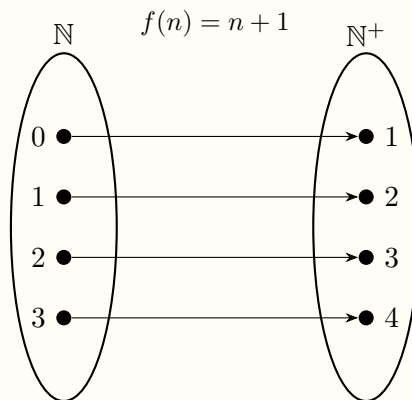
$$f(n) := n + 1.$$

Injective. Suppose $f(m) = f(n)$, i.e. $m + 1 = n + 1$. Then $m = n$.

Surjective. Let $y \in \mathbb{N}^+$, so $y \geq 1$. Then $y - 1 \in \mathbb{N}$ and $f(y - 1) = (y - 1) + 1 = y$.

Hence f is a bijection, so $|\mathbb{N}^+| = |\mathbb{N}|$. □

Remark. This already shows $\mathbb{N}$ is Dedekind infinite: $\mathbb{N}^+$ is a proper subset of $\mathbb{N}$ with $|\mathbb{N}^+| = |\mathbb{N}|$. Witnessing the equivalence theorem in the simplest possible setting.



6 Second example: $|\mathbb{Z}| = |\mathbb{N}|$

The integers stretch in two directions; $\mathbb{N}$ in only one. Yet they have the same size.

Theorem 6.1. $|\mathbb{Z}| = |\mathbb{N}|$.

Proof. Consider $f : \mathbb{Z} \rightarrow \mathbb{N}$ defined by

$$f(x) := \begin{cases} 2x & \text{if } x \geq 0, \\ -2x - 1 & \text{if } x < 0, \end{cases}$$

and $g : \mathbb{N} \rightarrow \mathbb{Z}$ defined by $g(x) := x$.

The map f sends non-negative integers to even naturals ($0 \mapsto 0, 1 \mapsto 2, 2 \mapsto 4, \dots$) and negative integers to odd naturals ($-1 \mapsto 1, -2 \mapsto 3, -3 \mapsto 5, \dots$). One can check f is a bijection: every natural number is reached exactly once, even numbers from the non-negative branch and odd numbers from the negative branch.

Hence $|\mathbb{Z}| = |\mathbb{N}|$. □

Remark. The map $g : \mathbb{N} \rightarrow \mathbb{Z}$ given by $g(x) = x$ is the obvious inclusion — it shows $\mathbb{N} \subseteq \mathbb{Z}$, hence $|\mathbb{N}| \leq |\mathbb{Z}|$. Combined with the bijection f , we get equality. Listing $\mathbb{Z}$ as

$$0, -1, 1, -2, 2, -3, 3, \dots$$

gives the same bijection in disguise — f^{-1} reads off the n -th term of this list.

$x \in \mathbb{Z}$	0	-1	1	-2	2	-3	3
$f(x) \in \mathbb{N}$	0	1	2	3	4	5	6

7 Hilbert’s hotel

The two theorems above are not isolated tricks — they are the discrete shadow of a single picture due to David Hilbert. The Grand Hotel is a thought experiment in which the strangeness of infinite cardinality becomes operational: rooms can be “made” for new guests in a fully booked hotel, simply by relabelling.

Remark (Setup). Imagine a hotel with rooms labelled by $\mathbb{N}^+ = \{1, 2, 3, \dots\}$ — one room for every positive integer. Suppose every room is occupied: for each $n \in \mathbb{N}^+$, room n houses guest G_n .

Scenario 1: One new guest arrives

A traveller shows up at the front desk. With finitely many rooms, the manager would say “sorry, full.” In the Grand Hotel, instead:

Every guest G_n moves from room n to room $n + 1$.

After the shift, room 1 is empty. The new guest takes room 1. No one is left out, because the map $n \mapsto n + 1$ is a bijection $\mathbb{N}^+ \rightarrow \mathbb{N}^+ \setminus \{1\}$ — exactly the bijection underlying $|\mathbb{N}^+| = |\mathbb{N}|$ from §5.

Theorem 7.1 (One new guest). There is a bijection $\mathbb{N}^+ \rightarrow \mathbb{N}^+$ that “frees up” room 1, namely $n \mapsto n + 1$ as a map onto $\{2, 3, 4, \dots\}$.

Scenario 2: Countably many new guests arrive

Now a bus pulls up carrying guests $H_1, H_2, H_3, \dots$ — one for each positive integer.

Every guest G_n moves from room n to room $2n$.

After the shift, the even rooms $\{2, 4, 6, \dots\}$ are occupied by old guests, and the odd rooms $\{1, 3, 5, \dots\}$ are empty. Place new guest H_k in room $2k - 1$.

Remark. Old guests use the bijection $\mathbb{N}^+ \rightarrow 2\mathbb{N}^+$ ($n \mapsto 2n$); new guests use the bijection $\mathbb{N}^+ \rightarrow 2\mathbb{N}^+ - 1$ ($k \mapsto 2k - 1$). Together they exhibit a bijection

$$\mathbb{N}^+ \sqcup \mathbb{N}^+ \longrightarrow \mathbb{N}^+,$$

which is precisely the content of $|\mathbb{Z}| = |\mathbb{N}|$ from §6 in another guise (negative integers $\sim$ new

guests, non-negative integers $\sim$ old guests).

Scenario	Reassignment	Underlying bijection
1 new guest	$G_n \rightarrow \text{room } n + 1$	$n \mapsto n + 1$
Bus of $\mathbb{N}^+$ guests	$G_n \rightarrow \text{room } 2n, H_k \rightarrow \text{room } 2k - 1$	$\mathbb{N}^+ \sqcup \mathbb{N}^+ \rightarrow \mathbb{N}^+$

Caution. The hotel argument relies on *simultaneous* relabelling. You cannot move the guests one at a time and expect to reach a stable configuration — there is no “last” guest to move. The bijection is a single global act, not a process.

Remark (Why this matters). Hilbert’s hotel is the first hint that “infinite $+1 =$ infinite” and “infinite $+$ infinite $=$ infinite” — not as sloppy slogans but as honest cardinal equalities $|\mathbb{N}| + 1 = |\mathbb{N}|$ and $|\mathbb{N}| + |\mathbb{N}| = |\mathbb{N}|$. The Dedekind definition (§3) makes this exact: a set is infinite precisely when it admits such a self-rearrangement.

8 Summary

1. $\binom{n}{k}$ counts the k -subsets of an n -set: $\binom{n}{k} = |\{Z \in \mathcal{P}(X) : |Z| = k\}|$ when $|X| = n$.
2. X is **infinite** iff $|X|$ is not equal to $|[n]|$ for any $n \in \mathbb{N}$.
3. X is **Dedekind infinite** iff some proper subset $Y \subsetneq X$ satisfies $|Y| = |X|$; **Dedekind finite** iff every proper subset is strictly smaller.
4. For finite A, B : $A \subsetneq B \Rightarrow |A| < |B|$ and $|A| \neq |B|$. Finite sets are Dedekind finite.
5. infinite $\iff$ Dedekind infinite (stated; full proof deferred).
6. $|\mathbb{N}^+| = |\mathbb{N}|$ via $f(n) = n + 1$.
7. $|\mathbb{Z}| = |\mathbb{N}|$ via $f(x) = 2x$ for $x \geq 0$, $f(x) = -2x - 1$ for $x < 0$.
8. Hilbert’s hotel concretizes both bijections: shift by one to make room for a guest, double up to make room for a bus.

Remark (Looking ahead). We have only met one “size of infinity” so far — the cardinality of $\mathbb{N}$, called *countable*. The next natural question is whether *every* infinite set has this size. Cantor’s diagonal argument will show that $|\mathbb{R}| > |\mathbb{N}|$, exposing a strict hierarchy of infinities.